

	$= 1.45 \times 10^{-18} \text{ V m}$		
5ai	More likely (higher probability) for electrons at the next higher energy level to de-excite to fill up the vacancy in the K shell.	A	A1
5aia	Due to the <u>higher atomic number</u> , the <u>attraction</u> between the nucleus and the inner shell electrons is <u>stronger</u> OR the inner shell electrons of tungsten is <u>more stable</u> due to the <u>higher atomic number</u> At <u>low accelerating potentials</u> , the energy of electrons is not sufficient OR At <u>high enough accelerating potentials</u> , the energy of electrons is sufficient	D	B1 B1

	to <u>knock electrons out of the inner shells</u> of the tungsten atom.		
5b	$E = \frac{hc}{e\lambda}$ $= \frac{(6.63 \times 10^{-34})(3.0 \times 10^8)}{(1.6 \times 10^{-19})(6.6 \times 10^{-11})}$ $= 1.9 \times 10^4 \text{ eV}$	A	C1 A1
6ai	Change in mass Δm	A	